1.6.3 Digital Vernier caliper

We are living in a digital world and the digital version of the Vernier callipers are available nowadays. Digital Vernier caliper (Figure 1.5) has a digital display on the slider, which calculates and displays the measured value. The user need not manually calculate the least count, zero error etc.

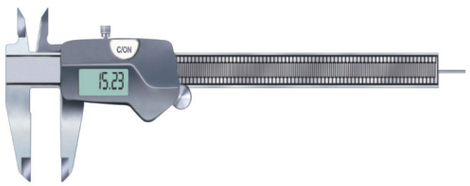


Figure 1.5 Digital Vernier caliper

1.7 Screw Gauge

Screw gauge is an instrument that can measure the dimensions upto $1/100^{th}$ of a millimetre or 0.01 mm. With the screw gauge it is possible to measure the diameter of a thin wire and thickness of thin metallic plates.

1.7.1 Description of screw gauge

The screw gauge consists of a U shaped metal frame. A hollow cylinder is attached to one end of the frame. Grooves are cut on the inner surface of the cylinder through which a screw passes (Figure 1.6). On the cylinder parallel to the axis of the screw there is a scale which is graduated in millimetre. It is called Pitch Scale (PS). One end of the screw is attached to a sleeve. The head of the sleeve (Thimble) is divided into 100 divisions and it is called the Head scale.

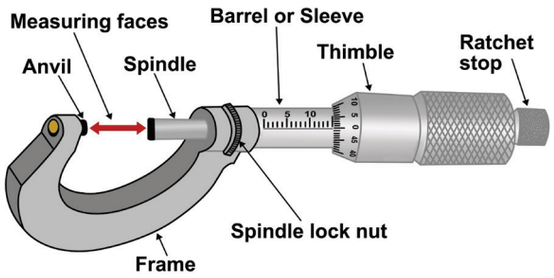


Figure 1.6 Screw gauge

The end of the screw has a plane surface (Spindle). A stud (Anvil) is attached to the other end of the frame, just opposite to the tip of the screw. The screw head is provided with a ratchat arrangement (safety device) to prevent the user from exerting undue pressure.

1.7.2 Using the screw gauge

The screw gauge works on the principal that when a screw rotates in a nut, the distance moved by the tip of the screw is directly proportional to the number of rotations.

a) Pitch of the screw

The pitch of the screw is the distance moved by the tip of the screw for one complete rotation of the head. It is equal to 1 mm in typical screw gauges.

$$\text{Pitch of the screw} = \frac{\text{Distance moved by the Pitch}}{\text{No. of rotations by Head scale}}$$

b) Least count of a screw gauge

The distance moved by the tip of the screw for a rotation of one division on the head scale is called the least count of the screw gauge.

$$\text{Least count of the instrument (L.C.)} = \frac{\text{Value of one smallest pitch scale reading}}{\text{Total number of Head scale division}}$$

$$\text{LC} = \frac{1}{100} = 0.01 \text{ mm}$$

c) Zero Error of a screw gauge

When the movable stud of the screw and the opposite fixed stud on the frame are brought into contact, if the zero of the head scale coincides with the pitch scale axis there is no zero error.

Positive zero error

When the movable stud of the screw and the opposite fixed stud on the frame are brought into contact, if the zero of the head scale lies below the pitch scale axis, the zero error is

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